



TRACKIFY:
DIGITAL PLATFORM FOR STUDENT ASSIGNMENT AND
PROJECT MANAGEMENT

Phase 2 - Information System Gathering and Requirement

Group Mario

Section 03

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SECD2613 System Analysis and Design

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	Content	Page
	1.0 Overview of The Project	3
	2.0 Problem Statement 2.1 Busy Schedules and Time Management Issues 2.2 Communication Gaps 2.3 Missed Deadlines 2.4 Difficulty with E-Learning Systems 2.5 Lack of Assignments Progress Monitoring	3
	3.0 Proposed Solutions 3.1 Smart Reminders 3.2 Assignment Progress Tracking 3.3 Group Assignment Tracking 3.4 Priority Ranking System Mode	4
	4.0 Information Gathering Process 4.1 Method Used 4.2 Summary from Method Used	4 - 9
	5.0 Requirement Analysis 5.1 Current Business Process 5.2 Functional Requirement 5.3 Non-functional Requirement 5.4 Logical DFD AS-IS System	10 - 13
	6.0 Summary of Requirement Analysis Process	14
	7.0 GitHub Information	14

1.0 Overview of The Project

The aim of this project is to develop tool that helps students manage assignments and group projects more effectively. The system is designed to assist students in organizing tasks, monitoring assignment progress, and keeping track of deadlines in one centralized platform. Currently, students often rely on multiple platforms such as e-learning systems, notes, and messaging applications, which may cause scattered information and missed deadlines.

Trackify is proposed to improve assignment management by providing features such as smart reminders, assignment progress tracking, group collaboration tools, and a priority ranking system. These features can help students manage their academic responsibilities more efficiently while improving communication and coordination among group members. The system also aims to reduce stress, improve time management, and support students in maintaining better academic performance.

2.0 Problem Statement

2.1 Busy Schedules and Time Management Issues

Packed timetables and involvement in university programs or associations are challenging for students to allocate time effectively.

2.2 Communication Gaps

For group projects, there may be lack of communication between the team members regarding project progress for divided tasks.

2.3 Missed Deadlines

Students may forget assignment deadlines or struggle with tracking multiple assignments from different classes.

2.4 Difficulty with E-Learning Systems

The system only shows the deadlines when students are logged in which makes it difficult and inconvenience in some cases. It also does not provide a progress tracker for progress monitoring.

2.5 Lack of Assignments Progress Monitoring

Students lack proper tools to visualize progress and track task completion.

3.0 Proposed Solutions

3.1 Smart Reminders

Popped up notification on the user's smartphone or tablet. Student does not have to login into e-learning and find the assignment deadlines for each subject manually.

3.2 Assignment Progress Tracking

Displaying the percentages of work done with a to do list.

3.3 Group Assignment Tracking

Team members will be able to communicate with each other more efficiently and see their real time progress.

3.4 Priority Ranking System Mode

Assignment or project with the highest priority, whether it be the closest due date and such, will automatically move to the top of the assignment list.

4.0 Information Gathering Process

4.1 Method Used

The questionnaire method was used to gather information and understand the challenges faced by students in managing assignments and project deadlines. This method was designed using a pyramid structure, beginning with closed-ended questions and moving towards open-ended questions to collect feedback students' current assignment management methods, common difficulties, and expectations for the proposed system that was distributed among UTM students.

This method was chosen because it allows data to be collected from multiple respondents efficiently and helps identify common issues experienced by students. The responses obtained were then analyzed to support the requirement analysis process and determine suitable features for the Trackify system.

4.2 Summary from Method Used

From the questionnaire responses, several common challenges were identified. Students reported difficulties in tracking multiple deadlines, forgetting submission dates due to lack of reminders, and facing issues with group coordination. Respondents also suggested features such as a smart reminders, progress tracking, and group collaboration tools to improve assignment management.

4.2.1 Questionnaire

Assignment Tracker Website Survey

Hi everyone, we are students from System Analysis and Design (SECD2613) section 03. We are making a website named Trackify or known as an assignment tracker. We would like to seek your help to fill in this survey of a website which involves student's difficulties in managing assignments.

Gender *

- ☐ Male
- ☐ Female

Faculty *

- ☐ Faculty of Built Environment and Surveying
- ☐ Faculty of Chemical and Energy Engineering
- ☐ Faculty of Computing
- ☐ Faculty of Educational Sciences and Technology
- ☐ Faculty of Electrical Engineering
- ☐ Faculty of Management
- ☐ Faculty of Mechanical Engineering
- ☐ Faculty of Civil Engineering
- ☐ Faculty of Science
- ☐ Faculty of Social Sciences and Humanities

4.2.2 Close-ended Question

Age *

- ☐ 19-25
- ☐ 26-30
- ☐ 31 and above

Year *

- ☐ Year 1
- ☐ Year 2
- ☐ Year 3
- ☐ Year 4

In your opinion, is an Assignment Tracker needed? *

- ☐ Yes
- ☐ No

How do you usually keep track of your assignments? *

- ☐ Lecturer / Friends
- ☐ E-learning
- ☐ Set reminder in calendar
- ☐ Notes
- ☐ I don't keep track my assignments

How often do you miss your assignment's deadlines? *

- ☐ Never
- ☐ Rarely
- ☐ Often

What difficulties do you face in managing assignments? *

- ☐ Forgot deadlines
- ☐ Busy schedule
- ☐ Too many assignments
- ☐ Lack of reminders
- ☐ Poor time management

How like are you to use this system if developed? *

- ☐ Never
- ☐ Sometimes
- ☐ Often

4.2.3 Open-ended Question

What do you dislike about your current method of managing assignments? *

Your answer

What features would help you to manage assignment better? (Suggestion) *

Your answer

4.2.4 Survey Result

Gender	Faculty	Age	Year	In your opinion, is an Assignment Tracker needed?	How do you usually keep track of your assignments?	How often do you miss your assignment's deadlines?	What difficulties do you face in managing assignments?	How like are you to use this system if developed?	What do you dislike about your current method of managing assignments?	What features would help you to manage assignment better? (Suggestion)
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar	Never	Busy schedule, Too many assignments	Often	many assignment due date	add priority ranking based on due date
Male	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning	Never	Busy schedule, Too many assignments, Lack of reminders	Often	hard to track assignment due dates (which one need to be submitted first)	add priority assignment list
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders, Poor time management	Often	free rider exist	group chat and track group member progress
Female	Faculty of Computing	19-25	Year 1	Yes	E-learning, Set reminder in calendar, Notes	Never	Forgot deadlines, Busy schedule, Too many assignments	Often	forget deadlines	add reminder about deadline
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders	Often	hard to track assignment progress	put visual percentage to track assignment progress
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning	Rarely	Forgot deadlines, Busy schedule, Lack of reminders	Often	group contributions not tracked properly	add task sharing for each member
Male	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar	Never	Busy schedule, Too many assignments	Often	hard to monitor divided tasks progress	put status for each divided tasks
Male	Faculty of Computing	19-25	Year 1	Yes	E-learning, Set reminder in calendar, Notes	Never	Busy schedule, Too many assignments	Often	hard to discuss with member through multiple apps	built in chat for group assignment
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar, Notes	Never	Forgot deadlines, Busy schedule, Too many assignments	Often	to many duplicate assignment due dates	add priority ranking
Female	Faculty of Computing	19-25	Year 1	Yes	E-learning, Notes	Never	Busy schedule, Too many assignments	Often	busy schedule	add alert reminder

Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	hard to track group member progress	add updated group task
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning	Rarely	Busy schedule, Too many assignments	Often	forgot deadline / too many deadline	add reminder
Female	Faculty of Computing	19-25	Year 1	Yes	E-learning	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders, Poor time management	Often	lack of time management	add due date ranking
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning	Rarely	Busy schedule, Too many assignments, Lack of reminders	Often	hard to track group member task progress	add task status
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders, Poor time management	Often	forget deadline	add reminder
Male	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	many assignment and project	put priority due date
Female	Faculty of Computing	19-25	Year 1	Yes	E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	have busy schedule / many programs	add reminder to inform due date
Male	Faculty of Computing	19-25	Year 1	Yes	E-learning	Rarely	Forgot deadlines, Too many assignments	Often	poor communication in group project/assignment	notification for group chat update
Male	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar	Rarely	Busy schedule, Too many assignments, Lack of reminders	Often	hard to see assignment progress	add progress bar
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar	Rarely	Forgot deadlines, Busy schedule, Lack of reminders	Often	hard to monitor task completion	add to-do list for each assignment
Male	Faculty of Computing	19-25	Year 1	Yes	E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders, Poor time management	Often	hard to see urgent assignment	add priority ranking mode

Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	busy schedule	put alert reminder
Male	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders	Often	multiple deadline	sorted it by deadlines
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	hard to see assignment progress	add visual progress bar
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Poor time management	Often	hard to track assignment statement (which one is not complete)	put to do list
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	multiple deadline	priority ranking mode
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, Set reminder in calendar	Rarely	Forgot deadlines, Busy schedule, Too many assignments	Often	do not have reminder about deadline	add smart reminder
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders	Often	hard to communicate with group member	add built in chat with notification
Female	Faculty of Computing	19-25	Year 1	Yes	Lecturer / Friends, E-learning, Set reminder in calendar, Notes	Rarely	Forgot deadlines, Busy schedule, Too many assignments, Lack of reminders	Often	forget deadline	put reminder about deadline

5.0 Requirement Analysis

5.1 Current Business Process

Based on the survey conducted, students currently use several methods to keep track of assignment deadlines, including:

- Lecturer or friends
- E-learning platforms
- Personal notes
- Calendar reminders
- Email notification from graduate.utm.my email

Lecturer will usually inform the students when the submission for assignment is open in e-learning with the deadline, while e-learning will give students reminder on the dashboard section when the due is less than a week. The calendar on the e-learning marks some of the important assignments date but sometimes it can miss some assignment. An email notification from student's graduate email will pop up a few days before the deadline and after the student submitted their work. Some students also write their assignments deadlines on their notebook or smartphone. For group assignments or project, student discusses task in group chat (usually via Telegram or WhatsApp) and the tasks are distributed through messages.

Workflow:

1. Lecturer announces assignment and deadline.
2. Student checks the assignment details on e-learning or saves in their personal notes.
3. Email notification pops up few days before the deadline.
4. For group work, student discusses task in group chat.
5. Tasks are distributed through messages.
6. Progress updates are shared informally in chat.
7. Students ask and remind each other about deadlines.

5.2 Functional Requirement

5.2.1 User Management

- Students can register and log in/out the system.
System maintains user accounts with secure authentication.

5.2.2 Assignment Management

- Students can create, update, and delete assignments records.
- The system stores assignment details such as title, course, and deadline.

5.2.3 Priority Ranking Mode

- Assignments are automatically sorted by deadlines.

5.2.4 Smart Reminder System

- Automatic notifications are displayed on mobile devices for upcoming deadlines.
- The system prioritizes reminders based on due dates.

5.2.5 Assignment Progress Tracking

- Students can update assignment completion percentage.
- A visual progress bar is displayed for each assignment.

5.2.6 Group Collaboration

- Group members can share updates, comments, and files.
- The system displays individual contributions and overall group progress.

5.3 Non-functional Requirement

5.3.1 Performance

The system should load assignment lists within 2-3 seconds.

5.3.2 Usability

The system should provide a simple and user-friendly.

5.3.3 Reliability

The system should operate minimal downtime.

5.3.4 Accessibility

The system should be accessible via desktop or web browser.

5.3.5 Security

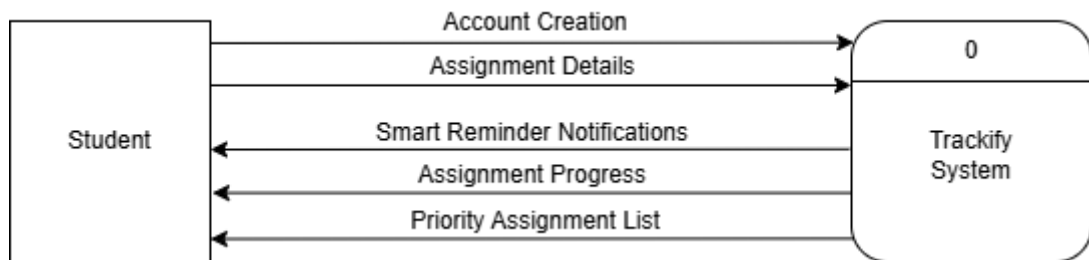
User authentication with encryption passwords.

5.3.6 Maintenance

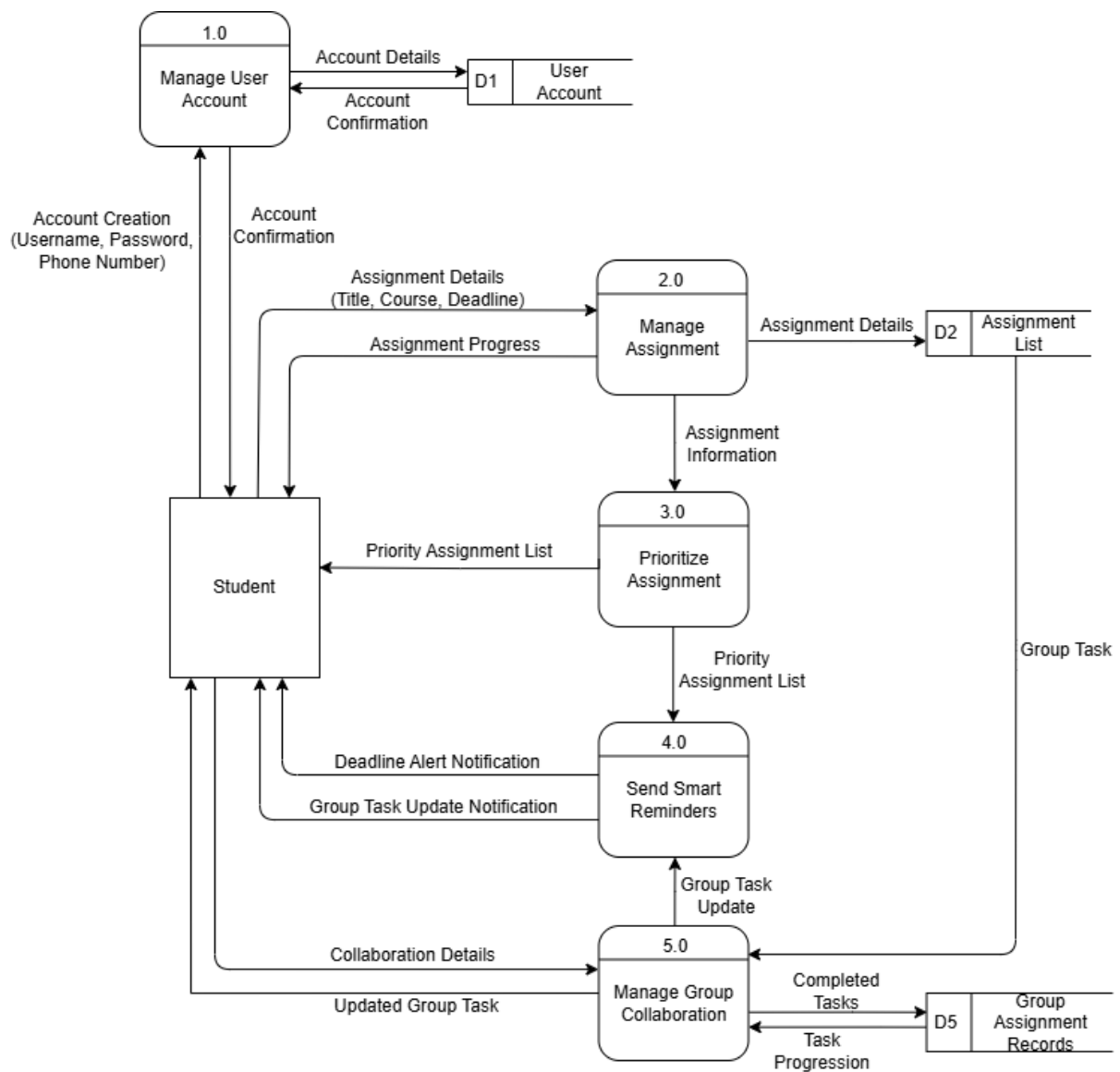
The system should support future updates easily.

5.4 Logical DFD AS-IS System

5.4.1 Context Diagram

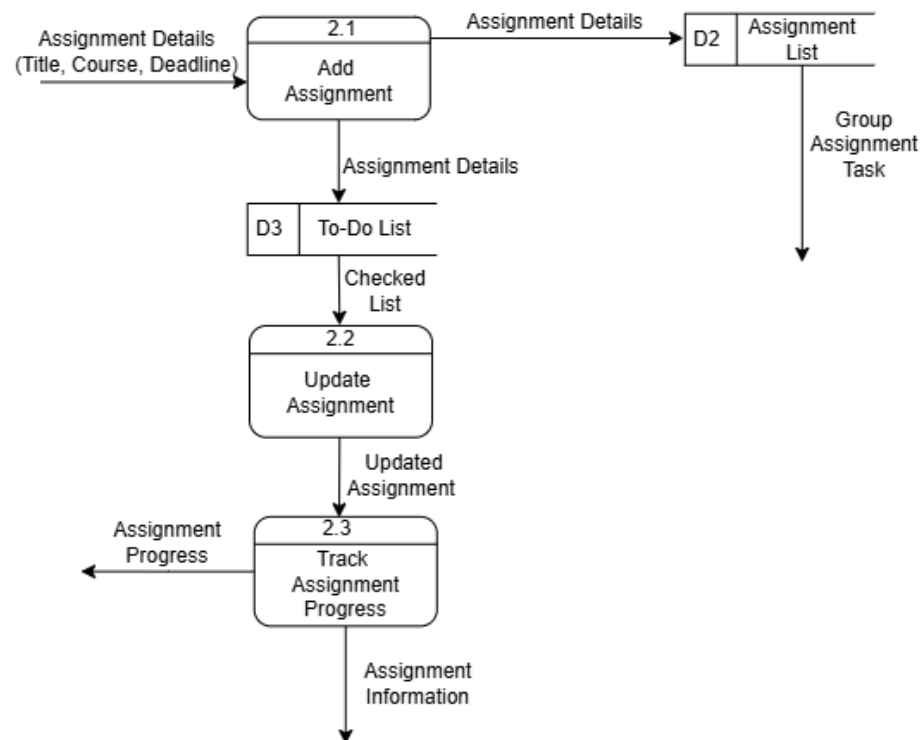


5.4.2 Diagram 0

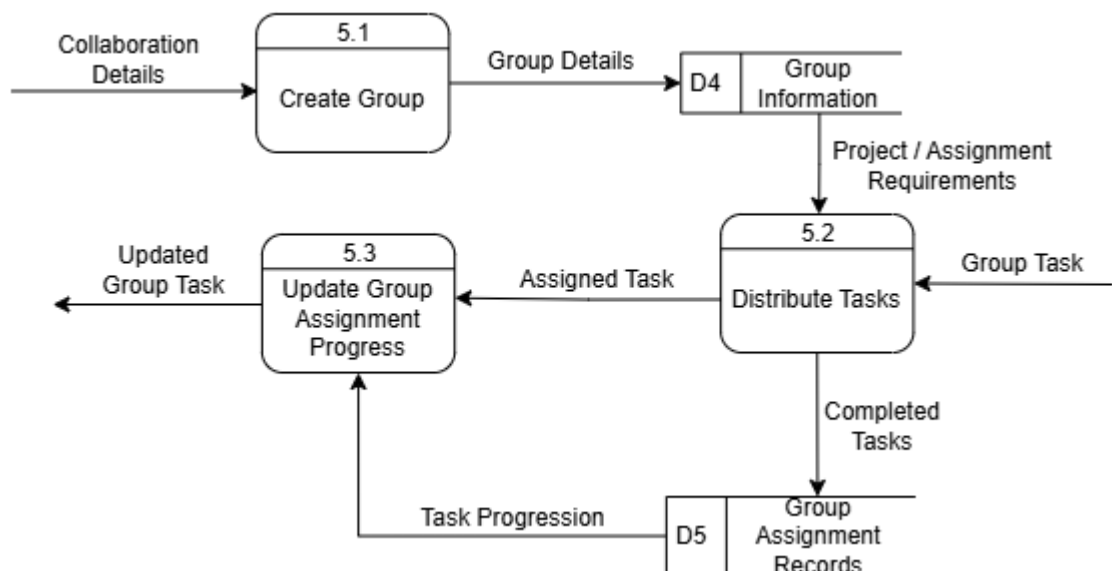


5.4.3 Child Diagram

5.4.3.1 Child Diagram Process 2.0 : Manage Assignment



5.4.3.2 Child Diagram Process 5.0 : Manage Group Collaboration



6.0 Summary of Requirement Analysis Process

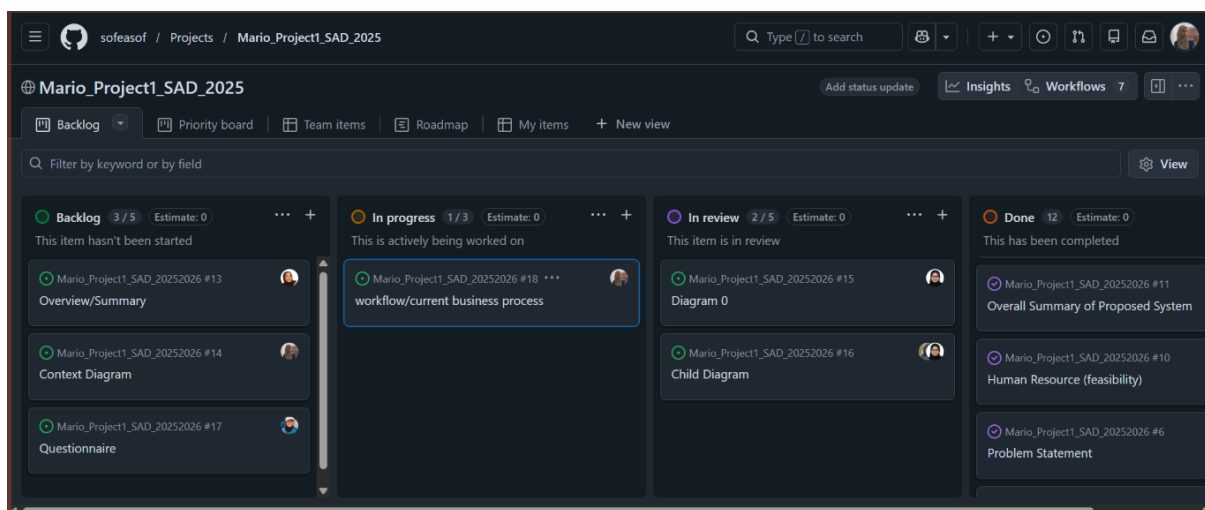
Trackify is designed as a centralized platform to help students manage assignments and group projects more effectively. By integrating smart reminders, progress tracking, group collaboration tools, and a priority ranking system, the platform addresses common issues such as missed deadlines, poor communication, and scattered information across multiple apps. The objective is to reduce stress, improve time management, and enhance academic performance by providing students with clear, organized view of their responsibilities.

The requirement analysis highlights both functional and non-functional needs, ensuring the system is practical and reliable. Students will be able to register, create and update assignments, track progress visually, and collaborate with peers in real time. At the same time, the system emphasizes usability, security, and accessibility, with features like fast loading times, encrypted authentication, and support for future updates. Overall, Trackify is developed to enhance efficiency, accountability, and seamless academic collaboration.

7.0 GitHub Information

URL:

- https://github.com/sofeasof/Mario_Project1_SAD_20252026
(Repository)
- <https://github.com/users/sofeasof/projects/1>
(Project)



Snapshot of the repository/project during phase 2.